

```
=====
TASK01
=====
```

```
SOURCE CODE
-----
```

```
Name: Tharun-Kumaran
Roll No: 2117250020478
```

```
package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478
```

```
/*Write a Java program that creates two boolean variables to represent whether it is
currently the weekend and whether it is raining. Then, print both values to the console.*/
```

```
public class task01 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        boolean whetherRaining=false;
        boolean weekend=true;
        System.out.println("Is whether raining?" + whetherRaining + "\nIs weekend ?" + weekend);

    }

}
```

```
OUTPUT
-----
```

```
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
Is whether raining?false
Is weekend ?true
```

```
=====
TASK02
=====
```

```
SOURCE CODE
-----
```

```
Name: Tharun-Kumaran
Roll No: 2117250020478
```

```
package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478
```

```
/*Write a Java program that uses byte variables to store small numbers representing a
smartphone's battery level and the day of the month.*/
```

```
public class task02 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        byte batteryLevel=90;
        System.out.println("Your battery level is:" + batteryLevel);

    }

}
```

```
OUTPUT
-----
```

```
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
```

Your battery level is:90

```
=====
TASK03
=====
```

```
SOURCE CODE
-----
```

```
Name: Tharun-Kumaran
Roll No: 2117250020478
```

```
package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Write a Java program that uses short variables to store moderate-sized numbers representing
geographic elevations (above and below sea level)*/
public class task03 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        short mtEverest = 8848;
        short marianaTrench = -10994;
        System.out.println("--Geographical locations--");
        System.out.println(" Mount Everest: " + mtEverest);
        System.out.println(" Marina Trench" + marianaTrench);

    }

}
```

```
OUTPUT
-----
```

```
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--Geographical locations--
Mount Everest: 8848
Marina Trench-10994
```

```
=====
TASK04
=====
```

```
SOURCE CODE
-----
```

```
Name: Tharun-Kumaran
Roll No: 2117250020478
```

```
package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Write a Java program that uses int variables to track a player's high score and their total
daily steps walked in a fitness mini-game.*/
public class task04 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        int dailySteps=4500;
        int highScore=7000;

        System.out.println("Your Steps Today: "+ dailySteps+"\nYour HighScore: "+highScore);

    }

}
```

OUTPUT

```
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
Your Steps Today: 4500
Your HighScore: 7000
```

TASK05

SOURCE CODE

```
Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create a class named Student with the following attributes:
â\200¢ Roll Number
â\200¢ Name
â\200¢ Department
â\200¢ CGPA
Create a constructor to initialize the values and display the student details using a method.*/

class Student{
    int rollNumber;
    String name;
    String department;
    float cgpa;

    public Student(int rollNumber,String name,String department,float cgpa) {
        this.rollNumber=rollNumber;
        this.name=name;
        this.department=department;
        this.cgpa=cgpa;
    }
    public int printRollNumber() {
        return rollNumber;
    }
    public String printName() {
        return name;
    }
    public String printDepartment() {
        return department;
    }
    public float printCgpa() {
        return cgpa;
    }
}

@Override
public String toString() {
    return "Name is: " + name
        + "\nAge, Department and CGPA are: "
        + rollNumber + " " + department + " " + cgpa;
}

}

public class task05 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Student s1= new Student(01,"Ram","CSE",9.0f);
        System.out.println(s1);

    }

}
```

OUTPUT

```
-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
Name is: Ram
Age, Department and CGPA are: 1 CSE 9.0
```

TASK06

SOURCE CODE

```
-----
Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create a Student class with attributes:
Roll Number , Name , Department
Create a method displayDetails() and invoke it using an object*/

class Student1{
    int rollNumber;
    String name;
    String department;

    public Student1(int rollNumber,String name,String department) {
        this.rollNumber=rollNumber;
        this.name=name;
        this.department=department;
    }

    public void displayDetails() {
        System.out.println("--Student Details");
        System.out.println("Roll Number: "+rollNumber);
        System.out.println("Name: "+name);
        System.out.println("Department: "+department);
    }

}

public class task06 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Student1 s1=new Student1(01,"Ram","CSE");
        s1.displayDetails();
    }

}
```

OUTPUT

```
-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--Student Details
Roll Number: 1
Name: Ram
Department: CSE
```

TASK07

SOURCE CODE

```

Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/* Employee Salary
Create an Employee class with:
Employee ID, Employee Name, Salary
Create a method showSalary() and call it using an object.*/

class Employee{
    int employeeID;
    String employeeName;
    String salary;

    public Employee(int employeeID,String employeeName,String salary) {
        this.employeeID=employeeID;
        this.employeeName=employeeName;
        this.salary=salary;
    }

    public void showSalary() {
        System.out.println("--Employee Details--");
        System.out.println("Employee ID: "+employeeID);
        System.out.println("Employee Name: "+ employeeName);
        System.out.println("Salary: "+salary);
    }
}

public class task07 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Employee e1= new Employee(01,"Ram","600000");
        e1.showSalary();
    }
}

```

OUTPUT

```

Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--Employee Details--
Employee ID: 1
Employee Name: Ram
Salary: 600000

```

TASK08

SOURCE CODE

```

Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create an abstract class Shape with an abstract method area().
Create:
    â\200¢ Rectangle
    â\200¢ Circle
Implement the area() method in both classes and display the area*/

```

```

abstract class Shape{
    public abstract double area();
}

class Rectangle extends Shape{
    private double length;
    private double width;

    public Rectangle(double length,double width) {
        this.length=length;
        this.width=width;
    }
    @Override
    public double area() {
        return length*width;
    }
}

class Circle extends Shape{
    private double r;

    public Circle(double r) {
        this.r=r;
    }
    @Override
    public double area() {
        return Math.PI*r*r;
    }
}

public class task08 {
    public static void main(String[] args) {
        Shape rectangle1=new Rectangle(1.1,2.2);
        Shape circle1=new Circle(2.0);

        System.out.printf("Area of the Rectangle: %.2f\n", rectangle1.area());
        System.out.printf("Area of the Circle: %.2f\n", circle1.area());

    }
}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Area of the Rectangle: 2.42
Area of the Circle: 12.57

```

TASK09

SOURCE CODE

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create an abstract class Employee with an abstract method calculateSalary().
Create:
    â\200¢ Manager
    â\200¢ Developer
Implement the salary calculation and display the salary*/

//Abstract base class
abstract class Employee3 {
    private String name;
    private int id;

    // Constructor

```

```

public Employee3(String name, int id) {
    this.name = name;
    this.id = id;
}

// Getters
public String getName() {
    return name;
}

public int getId() {
    return id;
}

// Abstract method to be implemented by subclasses
public abstract double calculateSalary();

// Concrete method to display employee details
public void displayEmployeeDetails() {
    System.out.println("-----");
    System.out.println("Employee ID    : " + id);
    System.out.println("Employee Name : " + name);
    System.out.printf("Calculated Pay: $%.2f%n", calculateSalary());
}
}

//Subclass Manager extending Employee
class Manager extends Employee3 {
    private double baseSalary;
    private double performanceBonus;

    public Manager(String name, int id, double baseSalary, double performanceBonus) {
        super(name, id);
        this.baseSalary = baseSalary;
        this.performanceBonus = performanceBonus;
    }

    // Implementing the abstract method
    @Override
    public double calculateSalary() {
        // Managers receive a base salary plus a performance bonus
        return baseSalary + performanceBonus;
    }
}

//Subclass Developer extending Employee
class Developer extends Employee3 {
    private double hourlyRate;
    private int hoursWorked;

    public Developer(String name, int id, double hourlyRate, int hoursWorked) {
        super(name, id);
        this.hourlyRate = hourlyRate;
        this.hoursWorked = hoursWorked;
    }

    // Implementing the abstract method
    @Override
    public double calculateSalary() {
        // Developers are paid based on hourly rate and hours worked
        return hourlyRate * hoursWorked;
    }
}

public class task09 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Employee3 manager = new Manager("Alice Johnson", 101, 7500.00, 1500.00);
        Employee3 developer = new Developer("Bob Smith", 102, 55.00, 160);

        // Displaying details and calculated salaries
        System.out.println("### Employee Payroll Breakdown ###");
        manager.displayEmployeeDetails();
        developer.displayEmployeeDetails();
    }
}

```

```
}
```

OUTPUT

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
### Employee Payroll Breakdown ###  
-----
```

```
Employee ID   : 101  
Employee Name : Alice Johnson  
Calculated Pay: $9000.00  
-----
```

```
Employee ID   : 102  
Employee Name : Bob Smith  
Calculated Pay: $8800.00
```

TASK10

SOURCE CODE

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478
```

```
package unit1;  
// Name: Tharun-Kumaran  
// Roll No: 2117250020478  
  
/*Create an abstract class Animal with an abstract method sound().  
Create:  
â\200¢ Dog  
â\200¢ Cat  
Display the sounds produced by each animal*/
```

```
//Define the abstract class Animal
```

```
abstract class Animal {  
    // Abstract method (does not have a body)  
    public abstract void sound();  
}
```

```
//Dog subclass extending Animal  
class Dog extends Animal {  
    @Override  
    public void sound() {  
        System.out.println("Bark");  
    }  
}
```

```
//Cat subclass extending Animal  
class Cat extends Animal {  
    @Override  
    public void sound() {  
        System.out.println("Meow");  
    }  
}
```

```
public class task10 {  
    public static void main(String[] args) {  
        System.out.println("Name: Tharun-Kumaran");  
        System.out.println("Roll No: 2117250020478");  
        // Create objects of Dog and Cat using Animal references  
        Animal myDog = new Dog();  
        Animal myCat = new Cat();  
  
        // Display the sounds produced by each animal  
        System.out.print("Dog says: ");  
        myDog.sound();  
  
        System.out.print("Cat says: ");  
        myCat.sound();  
    }  
}
```

OUTPUT

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
Dog says: Bark  
Cat says: Meow
```

TASK11

SOURCE CODE

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
  
package unit1;  
// Name: Tharun-Kumaran  
// Roll No: 2117250020478  
  
/*Create a class BankAccount with private variables:  
â\200¢ accountNumber  
â\200¢ balance  
Create getters and setters to access the data.*/  
  
public class task11 {  
    // Private variables (Data Hiding)  
    private String accountNumber;  
    private double balance;  
  
    // Optional: Constructor to initialize the object  
    public task11(String accountNumber, double initialBalance) {  
        this.accountNumber = accountNumber;  
        this.balance = initialBalance;  
    }  
  
    // Getter for accountNumber  
    public String getAccountNumber() {  
        return accountNumber;  
    }  
  
    // Setter for accountNumber  
    public void setAccountNumber(String accountNumber) {  
        this.accountNumber = accountNumber;  
    }  
  
    // Getter for balance  
    public double getBalance() {  
        return balance;  
    }  
  
    // Setter for balance  
    public void setBalance(double balance) {  
        // You can add validation logic here (e.g., balance cannot be negative)  
        if (balance >= 0) {  
            this.balance = balance;  
        } else {  
            System.out.println("Error: Balance cannot be negative.");  
        }  
    }  
}
```

OUTPUT

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
No console output.
```

TASK12

=====

SOURCE CODE

Name: Tharun-Kumaran
Roll No: 2117250020478

```
package unit1;  
// Name: Tharun-Kumaran  
// Roll No: 2117250020478
```

```
/*Create a class Employee with private variables:  
â\200¢ employeeID  
â\200¢ employeeName  
â\200¢ salary  
Use getter and setter methods to display the details.*/  
class Employee4{  
    private String employeeID;  
    private String employeeName;  
    private double salary;  
  
    public Employee4(String employeeID,String employeeName,double salary) {  
        this.employeeID=employeeID;  
        this.employeeName=employeeName;  
        this.salary=salary;  
    }  
    public String getEmployeeID() {  
        return employeeID;  
    }  
    public String getEmployeeName() {  
        return employeeName;  
    }  
    public double getSalary() {  
        return salary;  
    }  
  
    public void setEmployeeID(String employeeID) {  
        this.employeeID=employeeID;  
    }  
    public void setEmployeeName(String employeeName) {  
        this.employeeName=employeeName;  
    }  
    public void setSalary(double salary) {  
        this.salary=salary;  
    }  
}
```

```
public class task12 {  
  
    public static void main(String[] args) {  
        System.out.println("Name: Tharun-Kumaran");  
        System.out.println("Roll No: 2117250020478");  
        // TODO Auto-generated method stub  
        Employee4 e1=new Employee4("01","Ramu",60000);  
        System.out.println("Employee ID: "+e1.employeeID+" Employee Name: "+e1.employeeName+" Employee Salary  
: "+e1.salary);  
        e1.setSalary(80000);  
        System.out.println("Employee ID: "+e1.employeeID+" Employee Name: "+e1.employeeName+" Employee Salary  
: "+e1.salary);  
  
    }  
}}
```

OUTPUT

Name: Tharun-Kumaran
Roll No: 2117250020478
No console output.

=====

SOURCE CODE

Name: Tharun-Kumaran
Roll No: 2117250020478

```
package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create:
â\200¢ Parent Class: Vehicle
â\200¢ Child Class: Bike
Methods:
â\200¢ start()
â\200¢ ride()
Display both methods using inheritance.*/

abstract class Vehicle{
    void start() {
        System.out.println("Starting.....");
    }
    abstract void ride();
}
class Bike extends Vehicle{
    @Override
    void ride() {
        System.out.println("Riding.....");
    }
}
public class task13 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Vehicle v1=new Bike();
        v1.start();
        v1.ride();
    }

}
```

OUTPUT

Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
Starting.....
Riding.....

=====

TASK14

=====

SOURCE CODE

Name: Tharun-Kumaran
Roll No: 2117250020478

```
package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create an abstract class Vehicle with:
Abstract Method
start()
Subclasses
â\200¢ Car
â\200¢ Bike
Display how each vehicle starts.*/
```

```

abstract class Vehicle1 {
    // Abstract method (has no body)
    abstract void start();
}

// Subclass Car extending Vehicle
class Car extends Vehicle1 {
    @Override
    void start() {
        System.out.println("The car starts with a push-button ignition or turn-key.");
    }
}

// Subclass Bike extending Vehicle
class Bike1 extends Vehicle1 {
    @Override
    void start() {
        System.out.println("The bike starts with a kick-start or an electric button.");
    }
}

public class task14 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Vehicle1 myCar = new Car();
        Vehicle1 myBike = new Bike1();

        // Display how each vehicle starts
        myCar.start();
        myBike.start();

    }

}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
The car starts with a push-button ignition or turn-key.
The bike starts with a kick-start or an electric button.

```

TASK15

SOURCE CODE

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create a class Book with:
â\200¢ Book ID
â\200¢ Title
â\200¢ Author
Create an object and display the book details.*/

class Book {
    // Attributes (Instance Variables)
    private int bookId;
    private String title;
    private String author;

    // Parameterized Constructor to initialize the attributes
    public Book(int bookId, String title, String author) {

```

```

        this.bookId = bookId;
        this.title = title;
        this.author = author;
    }

    // Method to display the book details
    public void displayDetails() {
        System.out.println("--- Book Details ---");
        System.out.println("Book ID : " + bookId);
        System.out.println("Title   : " + title);
        System.out.println("Author  : " + author);
    }
}

public class task15 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Book myBook = new Book(101, "Effective Java", "Joshua Bloch");

        // Display the book details
        myBook.displayDetails();

    }

}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--- Book Details ---
Book ID : 101
Title   : Effective Java
Author  : Joshua Bloch

```

TASK16

SOURCE CODE

```

Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create an object for an Employee class and display:
â\200¢ Employee ID
â\200¢ Employee Name
â\200¢ Salary*/

class Employee5 {
    // Instance variables (Attributes)
    private int employeeId;
    private String employeeName;
    private double salary;

    // Parameterized constructor to initialize the variables
    public Employee5(int employeeId, String employeeName, double salary) {
        this.employeeId = employeeId;
        this.employeeName = employeeName;
        this.salary = salary;
    }

    // Method to display the employee details
    public void displayEmployeeInfo() {
        System.out.println("--- Employee Details ---");
        System.out.println("Employee ID   : " + employeeId);
        System.out.println("Employee Name : " + employeeName);
        System.out.println("Salary       : $" + salary);
    }
}

```

```

    }
}

public class task16 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Employee5 emp = new Employee5(101, "Alice Smith", 75000.50);

        // Display the details of the created object
        emp.displayEmployeeInfo();
    }

}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--- Employee Details ---
Employee ID   : 101
Employee Name : Alice Smith
Salary       : $75000.5

```

TASK17

SOURCE CODE

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create methods:
â\200¢ add()
â\200¢ subtract()
â\200¢ multiply()
Display the results using an object.*/

class Calculator {
    // Methods returning the results
    public double add(double a, double b) {
        return a + b;
    }

    public double subtract(double a, double b) {
        return a - b;
    }

    public double multiply(double a, double b) {
        return a * b;
    }
}

public class task17 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        Calculator calc = new Calculator();

        // Define input numbers
        double num1 = 20.0;
        double num2 = 5.0;

        // Display the results using the object
        System.out.println("Addition: " + calc.add(num1, num2));
    }
}

```

```

        System.out.println("Subtraction: " + calc.subtract(num1, num2));
        System.out.println("Multiplication: " + calc.multiply(num1, num2));
    }
}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
Addition: 25.0
Subtraction: 15.0
Multiplication: 100.0

```

TASK18

SOURCE CODE

```

Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Write a Java Program to demonstrate all four access specifiers using two classes in two differen*/

// Helper class within the same package to hold different access levels
class DataHolder {
    // 1. Public: Accessible from anywhere
    public String publicVar = "Public Variable: Accessible everywhere.";

    // 2. Protected: Accessible within the same package and by subclasses
    protected String protectedVar = "Protected Variable: Accessible in same package.";

    // 3. Default (No keyword): Accessible only within the same package
    String defaultVar = "Default Variable: Accessible within the package.";

    // 4. Private: Accessible ONLY inside this class
    private String privateVar = "Private Variable: Restrictive, inside DataHolder only.";

    // Public method to show all variables inside their own class
    public void displayAll() {
        System.out.println("--- Inside DataHolder Class ---");
        System.out.println(publicVar);
        System.out.println(protectedVar);
        System.out.println(defaultVar);
        System.out.println(privateVar); // Fully accessible here
    }

    // Getter method to allow controlled access to the private variable outside
    public String getPrivateVar() {
        return privateVar;
    }
}

public class task18 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // Instantiate the helper class
        DataHolder holder = new DataHolder();

        // Display access from within the helper class itself
        holder.displayAll();
        System.out.println();

        // Demonstrate access from a separate class (task18) in the same package
    }
}

```

```

        System.out.println("--- Accessing from task18 Class ---");

        // Accessing public, protected, and default variables directly
        System.out.println(holder.publicVar);
        System.out.println(holder.protectedVar);
        System.out.println(holder.defaultVar);

        // CRITICAL: Un-commenting the line below will throw a compile-time error
        // System.out.println(holder.privateVar);

        // Accessing the private variable safely using a public getter method
        System.out.println("Indirectly: " + holder.getPrivateVar());
    }
}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--- Inside DataHolder Class ---
Public Variable: Accessible everywhere.
Protected Variable: Accessible in same package.
Default Variable: Accessible within the package.
Private Variable: Restrictive, inside DataHolder only.

--- Accessing from task18 Class ---
Public Variable: Accessible everywhere.
Protected Variable: Accessible in same package.
Default Variable: Accessible within the package.
Indirectly: Private Variable: Restrictive, inside DataHolder only.

```

TASK19

SOURCE CODE

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Create a class Student with:
â\200¢ Roll Number
â\200¢ Name
â\200¢ Static variable college
Create two student objects and display their details.*/
class Student6 {
    // Instance variables (unique to each object)
    int rollNo;
    String name;

    // Static variable (shared among all objects)
    static String college = "RIT";

    // Constructor to initialize instance variables
    public Student6(int rollNo, String name) {
        this.rollNo = rollNo;
        this.name = name;
    }

    // Method to display student details
    public void displayDetails() {
        System.out.println("Roll Number: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("College: " + college);
        System.out.println("-----");
    }
}

public class task19 {

```

```

        public static void main(String[] args) {
            System.out.println("Name: Tharun-Kumaran");
            System.out.println("Roll No: 2117250020478");
            // TODO Auto-generated method stub
            Student6 student1 = new Student6(101, "Alice");
            Student6 student2 = new Student6(102, "Bob");

            // Displaying details of both students
            System.out.println("--- Student 1 Details ---");
            student1.displayDetails();

            System.out.println("--- Student 2 Details ---");
            student2.displayDetails();

        }
    }
}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--- Student 1 Details ---
Roll Number: 101
Name: Alice
College: RIT
-----
--- Student 2 Details ---
Roll Number: 102
Name: Bob
College: RIT
-----

```

TASK20

SOURCE CODE

```

Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Write a Java program to implement a Mobile Store using a default constructor. Create a Mobile class
with the attributes brand and price. The default constructor should display a welcome message when
a mobile object is created. Display the mobile details after object creation.*/
class Mobile {
    // Attributes
    String brand;
    double price;

    // Default constructor (no-argument constructor)
    Mobile() {
        System.out.println("Welcome to the Mobile Store!");
        brand = "Samsung";
        price = 75000.0;
    }

    // Method to display mobile details
    void displayDetails() {
        System.out.println("Mobile Brand: " + brand);
        System.out.println("Mobile Price: â\2021" + price);
    }
}

public class task20 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
    }
}

```

```

        // TODO Auto-generated method stub
        Mobile myMobile = new Mobile();

        System.out.println("\nDisplaying Mobile Details After Creation:");
        myMobile.displayDetails();

    }

}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
Welcome to the Mobile Store!

```

```

Displaying Mobile Details After Creation:
Mobile Brand: Samsung
Mobile Price: â\202175000.0

```

TASK21

SOURCE CODE

```

Name: Tharun-Kumaran
Roll No: 2117250020478

```

```

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

```

```

/*Write a Java program to implement a Mobile Store using a parameterized constructor. Create
a Mobile class with the attributes brand and price. Accept the values through the constructor and
display the details of two different mobile objects.*/

```

```

class Mobile1 {
    String brand;
    double price;

    // Parameterized constructor to initialize brand and price
    Mobile1(String brand, double price) {
        this.brand = brand;
        this.price = price;
    }

    // Method to display mobile details
    void displayDetails() {
        System.out.println("Brand: " + brand);
        System.out.println("Price: Rs " + price);
        System.out.println("-----");
    }
}

public class task21 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        System.out.println("=== Welcome to the Mobile Store ===");

        // Creating two different mobile objects using the parameterized constructor
        Mobile1 mobile1 = new Mobile1("Apple", 999.99);
        Mobile1 mobile2 = new Mobile1("Samsung", 799.50);

        // Displaying the details of both mobile objects
        mobile1.displayDetails();
        mobile2.displayDetails();

    }
}

```

```
}
```

OUTPUT

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
=== Welcome to the Mobile Store ===  
Brand: Apple  
Price: Rs 999.99  
-----  
Brand: Samsung  
Price: Rs 799.5  
-----
```

TASK22

SOURCE CODE

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
  
package unit1;  
// Name: Tharun-Kumaran  
// Roll No: 2117250020478  
  
/* Write a Java program to implement a user-defined copy constructor in a Mobile Store application.  
Create a mobile object with a brand and price, then create another object by copying the first object  
using the copy constructor. Display the details of both objects.*/  
  
class Mobile2 {  
    private String brand;  
    private double price;  
  
    // Parameterized constructor to initialize the first mobile object  
    public Mobile2(String brand, double price) {  
        this.brand = brand;  
        this.price = price;  
    }  
  
    // User-defined copy constructor  
    // It accepts an object of the same class as a parameter  
    public Mobile2(Mobile2 otherMobile) {  
        this.brand = otherMobile.brand; // Copying the brand  
        this.price = otherMobile.price; // Copying the price  
    }  
  
    // Method to display mobile details  
    public void displayDetails(String objectLabel) {  
        System.out.println(objectLabel + " -> Brand: " + brand + ", Price: $" + price);  
    }  
}  
  
public class task22 {  
  
    public static void main(String[] args) {  
        System.out.println("Name: Tharun-Kumaran");  
        System.out.println("Roll No: 2117250020478");  
        // TODO Auto-generated method stub  
        System.out.println("--- Mobile Store Application --- \n");  
  
        // 1. Create the original mobile object using the parameterized constructor  
        Mobile2 originalMobile = new Mobile2("Apple iPhone 15", 799.99);  
  
        // 2. Create a duplicate object by passing the original object into the copy constructor  
        Mobile2 copiedMobile = new Mobile2(originalMobile);  
  
        // 3. Display the details of both objects to verify they match  
        originalMobile.displayDetails("Original Mobile");  
        copiedMobile.displayDetails("Copied Mobile ");  
    }  
}  
}
```

OUTPUT

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
--- Mobile Store Application ---
```

```
Original Mobile -> Brand: Apple iPhone 15, Price: $799.99  
Copied Mobile   -> Brand: Apple iPhone 15, Price: $799.99
```

TASK23

SOURCE CODE

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
  
package unit1;  
// Name: Tharun-Kumaran  
// Roll No: 2117250020478  
  
/*Write a Java program to demonstrate a no-argument constructor using a Mobile Store application.  
Initialize the mobile details inside the constructor and display the details using a member function.*/  
  
class Mobile3 {  
    // Instance variables (Mobile details)  
    String brand;  
    String model;  
    double price;  
    int ramSize; // in GB  
  
    // No-argument constructor to initialize mobile details  
    Mobile3() {  
        brand = "Samsung";  
        model = "Galaxy S24";  
        price = 79999.00;  
        ramSize = 8;  
    }  
  
    // Member function to display mobile details  
    void displayDetails() {  
        System.out.println("--- Mobile Store Item Details ---");  
        System.out.println("Brand: " + brand);  
        System.out.println("Model: " + model);  
        System.out.println("Price: â\2021" + price);  
        System.out.println("RAM: " + ramSize + " GB");  
    }  
}  
  
public class task23 {  
  
    public static void main(String[] args) {  
        System.out.println("Name: Tharun-Kumaran");  
        System.out.println("Roll No: 2117250020478");  
        // TODO Auto-generated method stub  
        Mobile3 myMobile = new Mobile3();  
  
        // Calling the member function to display the initialized details  
        myMobile.displayDetails();  
    }  
}  
}
```

OUTPUT

```
-----  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
Name: Tharun-Kumaran  
Roll No: 2117250020478  
--- Mobile Store Item Details ---  
Brand: Samsung
```

Model: Galaxy S24
Price: ₹202,7999.0
RAM: 8 GB

=====

TASK24

=====

SOURCE CODE

Name: Tharun-Kumaran
Roll No: 2117250020478

```
package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Write a Java program to demonstrate constructor overloading in a Mobile Store
application. Create a Mobile class with three constructors:
â227\217 A default constructor
â227\217 A constructor that accepts only the mobile brand
â227\217 A constructor that accepts both the mobile brand and price
Create objects using all three constructors and display their details.*/

class Mobile4 {
    // Instance variables
    private String brand;
    private double price;

    // 1. Default constructor
    public Mobile4() {
        this.brand = "Unknown Brand";
        this.price = 0.0;
    }

    // 2. Constructor that accepts only the mobile brand
    public Mobile4(String brand) {
        this.brand = brand;
        this.price = 0.0; // Default price when not specified
    }

    // 3. Constructor that accepts both the mobile brand and price
    public Mobile4(String brand, double price) {
        this.brand = brand;
        this.price = price;
    }

    // Method to display the details of the mobile object
    public void displayDetails() {
        System.out.println("Mobile Brand: " + brand + " | Price: $" + price);
    }
}

public class task24 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub
        System.out.println("--- Welcome to the Mobile Store --- \n");

        // Creating an object using the default constructor
        Mobile4 mobile1 = new Mobile4();

        // Creating an object using the brand-only constructor
        Mobile4 mobile2 = new Mobile4("Apple iPhone");

        // Creating an object using the brand and price constructor
        Mobile4 mobile3 = new Mobile4("Samsung Galaxy", 1199.99);

        // Displaying details of all created objects
        System.out.println("Object 1 Details (Default):");
        mobile1.displayDetails();

        System.out.println("\nObject 2 Details (Brand Only):");
        mobile2.displayDetails();
    }
}
```

```

        System.out.println("\nObject 3 Details (Brand and Price):");
        mobile3.displayDetails();

    }

}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--- Welcome to the Mobile Store ---

Object 1 Details (Default):
Mobile Brand: Unknown Brand | Price: $0.0

Object 2 Details (Brand Only):
Mobile Brand: Apple iPhone | Price: $0.0

Object 3 Details (Brand and Price):
Mobile Brand: Samsung Galaxy | Price: $1199.99

```

TASK25

SOURCE CODE

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478

package unit1;
// Name: Tharun-Kumaran
// Roll No: 2117250020478

/*Write a Java program that uses long variables to store massive numbers representing
galactic distances and global web searches.*/

public class task25 {

    public static void main(String[] args) {
        System.out.println("Name: Tharun-Kumaran");
        System.out.println("Roll No: 2117250020478");
        // TODO Auto-generated method stub

        long distanceToAndromedaKm = 2400000000000000000L;

        // Global web searches: Estimated total Google searches per year worldwide
        long annualGlobalWebSearches = 20000000000000L;

        // Print the values with descriptive text
        System.out.println("--- Galactic Distances ---");
        System.out.println("Distance to the Andromeda Galaxy: " + distanceToAndromedaKm + " km");

        System.out.println("\n--- Global Web Metrics ---");
        System.out.println("Estimated annual global web searches: " + annualGlobalWebSearches);

    }

}

```

OUTPUT

```

-----
Name: Tharun-Kumaran
Roll No: 2117250020478
Name: Tharun-Kumaran
Roll No: 2117250020478
--- Galactic Distances ---

```

Distance to the Andromeda Galaxy: 240000000000000000 km

--- Global Web Metrics ---

Estimated annual global web searches: 2000000000000